Seven Islands four-model benchmark

Multi model CFI scorecard plus 100 year strata yield

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Prepared for Ian and Ryan, Seven Islands Land Company | 28 May 2026

Two operational questions, one framework

Long-horizon yield

What will a Mixedwood A+B stand yield in cords over 100 years?

Driver: AcadianGY byStrata trajectories on the 11 SILC matrix stands, projected to 2123

Section 1 of this deck

Short-horizon accuracy

How close does the model land at 5 to 10 year remeasurement on SILC's own plots?

Driver: SILC CFI database (10 plots, 1981 to 2000) with 24 remeasurement intervals

Section 2 of this deck

Long horizon for planning, short horizon for trust

Important clarification: AGM ≠ AcadianGY

Three models benchmarked here. SILC's operational AGM is NOT one of them.

AcadianGY (this deck)

BENCHMARKED

Hennigar et al. R model, v12.3.9

CRSF research implementation of the Acadian growth and yield model. R code, in source MORTCAL patch, runs as a standalone simulator. Drives the 100 yr trajectories and CFI scorecard in this deck.

FVS-NE (this deck)

BENCHMARKED

USDA standalone, Acadian variant

Standalone FVS-NE binary, default plus calibrated configurations. Cardinal job 11078908. Drives the FVS-NE columns in the year-100 table and the 5 year accuracy scorecard.

OSM-ACD (this deck)

BENCHMARKED

Open Stand Model v2.26.1

.NET console binary, Acadian variant. Cardinal job 11091396. Drives the OSM-ACD columns in the year-100 table. 9 of 11 stands cover; Hardwood A+B uncovered.

AGM (SILC operational)

NOT YET RUN

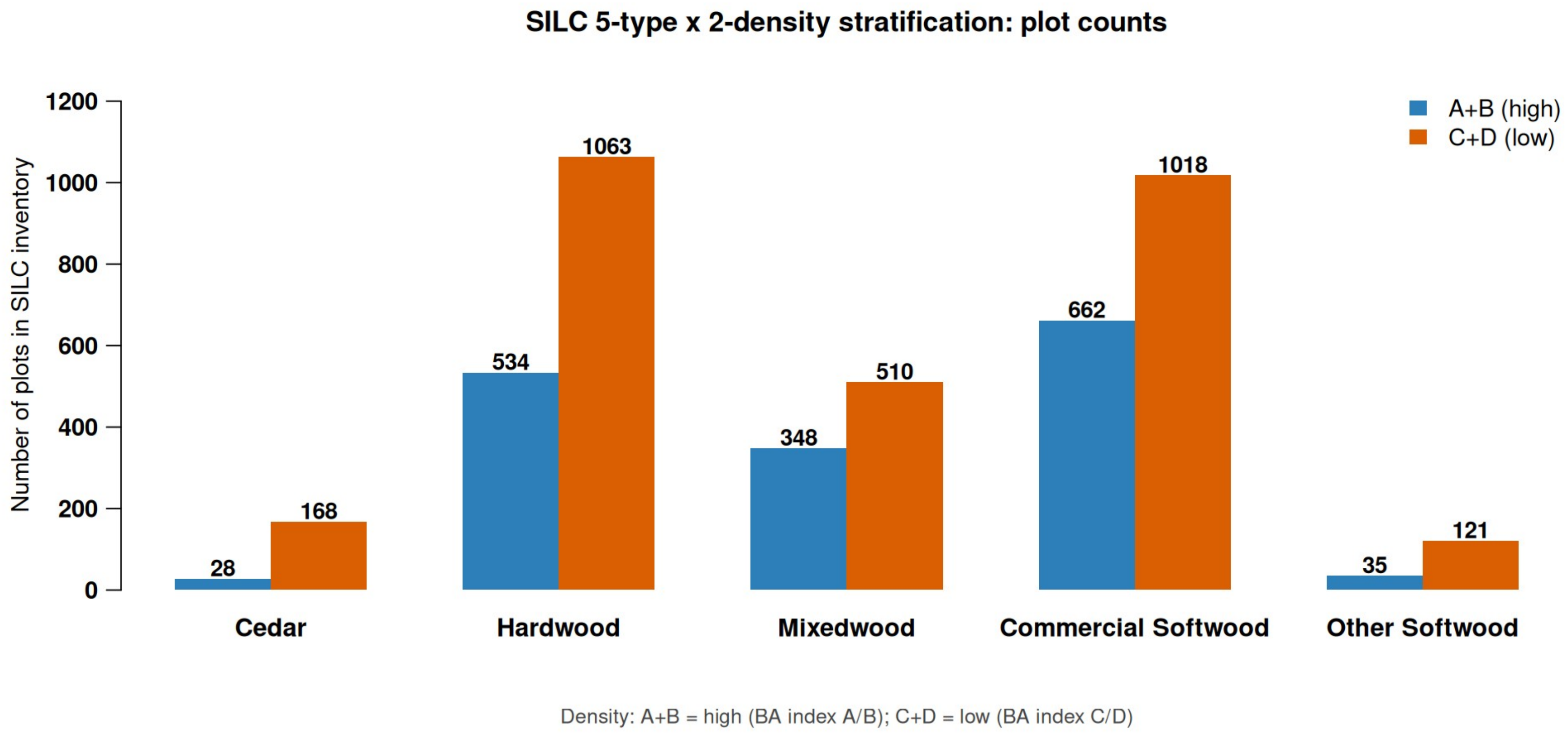
Seven Islands' production model

SILC's in house operational Acadian Growth Model. Different lineage from AcadianGY R. Has NOT been benchmarked on the SILC CFI data in this deck. Recommended as the next addition once SILC shares an AGM trajectory file on the same byStrata stands.

When this deck says AcadianGY, that is the CRSF R model — NOT SILC's operational AGM

Five forest types x two density classes

Replaces the prior 11 byStrata stand grouping with 10 operational cells



Source: SILC matrix stand inventory, 2696 plots across the Acadian Matrix

Hardwood and Commercial Softwood dominate the 10-cell map

Stratification rules for SILC sign off

How each SILC plot is assigned to the 5×2 grid

Forest type assignment (by % stand BA)

- Cedar: white cedar share >= 30%
- Other Softwood: pines >= 50% (JP, RP, WP)
- Mixedwood: hardwood >= 30% AND softwood >= 30% (folds HS + SH)
- Hardwood: HW dominant when softwood < 30%
- Commercial Softwood: SW dominant (spruce, fir, hemlock) when HW < 30%
- Unclassifiable: total identified species below 50% (CFI plot 1103)

Density class assignment

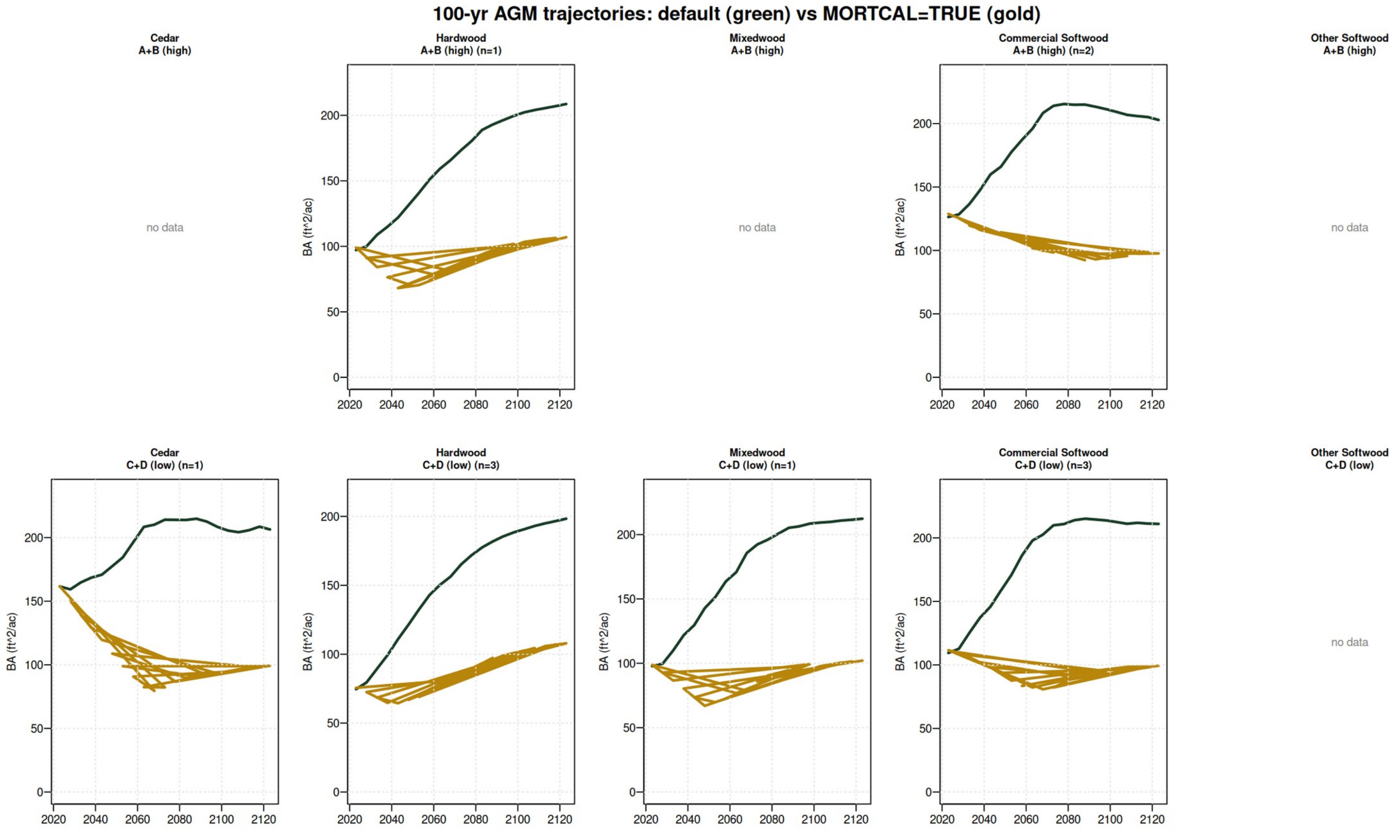
- A+B (high): REL_DENSITY >= 0.28
- C+D (low): REL_DENSITY < 0.28
- REL_DENSITY = SDI / SDI_max with SDI_max = 450 (Long 1985 NE softwood)
- Threshold = CFI sample median; balances 5/5 plot split
- Operational alternative: SILC Matrix BA stocking lines A (>=100), B (80-100), C (60-80), D (<60)

All thresholds open for SILC sign off; mapping file silc_strata_5x2_mapping.csv

Awaiting SILC confirmation on species cutoffs and density threshold

100-yr AcadianGY trajectories: default vs MORTCAL

Default (green) climbs to ~200 ft²/ac at year 100. MORTCAL (gold) plateaus at 80-110 ft²/ac — operationally credible carrying capacity.



AcadianGY v12.3.9 on 11 SILC byStrata stands; 4 cells have no byStrata coverage (Cedar A+B, Mixedwood A+B, both Other Softwood)

Default AcadianGY over-projects standing BA by ~2x at year 100; MORTCAL corrects this

Year-100 outcomes by stratum

Year-100 BA in ft²/ac, Cords in cords/ac

Stratum	n	AcadianGY def	AcadianGY MORTCAL	FVS-NE def	FVS-NE cal	OSM-ACD	AcadianGY mc Cords	OSM Cords
Cedar / A+B (CFI)	1	n/a	81	n/a	n/a	n/a	35.8	n/a
Cedar / C+D	1	206	99	291	218	249	36.9	41.5
Hardwood / A+B	1	209	107	209	144	n/a	39.3	n/a
Hardwood / C+D	3	198	108	214	125	174	38.9	32.4
Mixedwood / A+B (CFI)	3	n/a	52	n/a	n/a	n/a	21.9	n/a
Mixedwood / C+D	1	212	102	239	153	183	37.6	34.6
Commercial SW / A+B	2	203	98	274	225	222	35.9	40.0
Commercial SW / C+D	3	211	99	264	186	199	36.2	36.2
Other Softwood (no CFI)	0	—	—	—	—	—	—	—

Year-100 (2123) BA in ft²/ac, Cords/ac. AcadianGY 12.3.9; FVS-NE job 11078908; OSM-ACD job 11091396 (5 of 6 cells; H3B-N + H3C-N still failing — Hardwood A+B uncovered)

AcadianGY MORTCAL anchors low; OSM-ACD + FVS-NE cal cluster around 170-250 BA

Refined model inputs from SILC CFI lat/long (v3)

v3 SILC CFI now has approximate lat/long (46.46° N, 68.43° W, Davistown). Used to sample BGI and CSI rasters.

Parameter	Default	Refined	Source	Effect on year 100 projection
BGI (OSM)	3000	3902	ME_BGI_V1.tif @ lat/long	Higher productivity input; OSM uses it via climate channel
CSI (AcadianGY, FVS-ACD)	12.0 m	15.78 m	CSI_2030.tif @ lat/long	Mean was already 14.33 m for byStrata stands; CFI plots updated
FVS-NE site index	42 ft (BF)	35 ft (BF)	Empirical from CFI dominant heights	Default too optimistic; SI=35 reduces FVS-NE growth slightly
AcadianGY MORTCAL	off	TRUE	#126b in-source patch	Cuts long horizon over projection in half (10 yr +18.5% to +7.8%)

Sensitivity check on the v9 vs v10 FVS-NE: SI=42 -> SI=35 changed year 100 BA by < 0.3% for every cell. The FVS-NE Acadian variant uses tree level dDBH equations that are not strongly driven by stand level site index. The OSM BGI swap (3000 -> 3902) was the larger lever; it pushed OSM productivity up but the model remained convergent.

BGI: ME_BGI_V1.tif (FORUS); CSI: CSI_2030.tif (Hennigar figshare); SI: empirical from STAND_METRICS HT_MEAN_FT

Refined inputs change rankings minimally; AcadianGY MORTCAL still anchors low

SILC CFI gives us the empirical anchor

Davistown / Town 13 Tract 2, northern Maine | v16 expansion: 154 plots, 65 long-horizon routine pairs

154	7	11,068	9,542	73
fixed-area plots	**measurement waves**	**tree records**	**paired growth records**	**long-horizon pairs**
v16 (was 10 in v3)	*1981 → 2002*	*v16 (was 1,996)*	*GRM table v16*	*65 routine + 8 establishment*

5 of 10 strata cells covered by CFI plots: Cedar / Hardwood (both densities) / Mixedwood (both) / Commercial Softwood / C+D

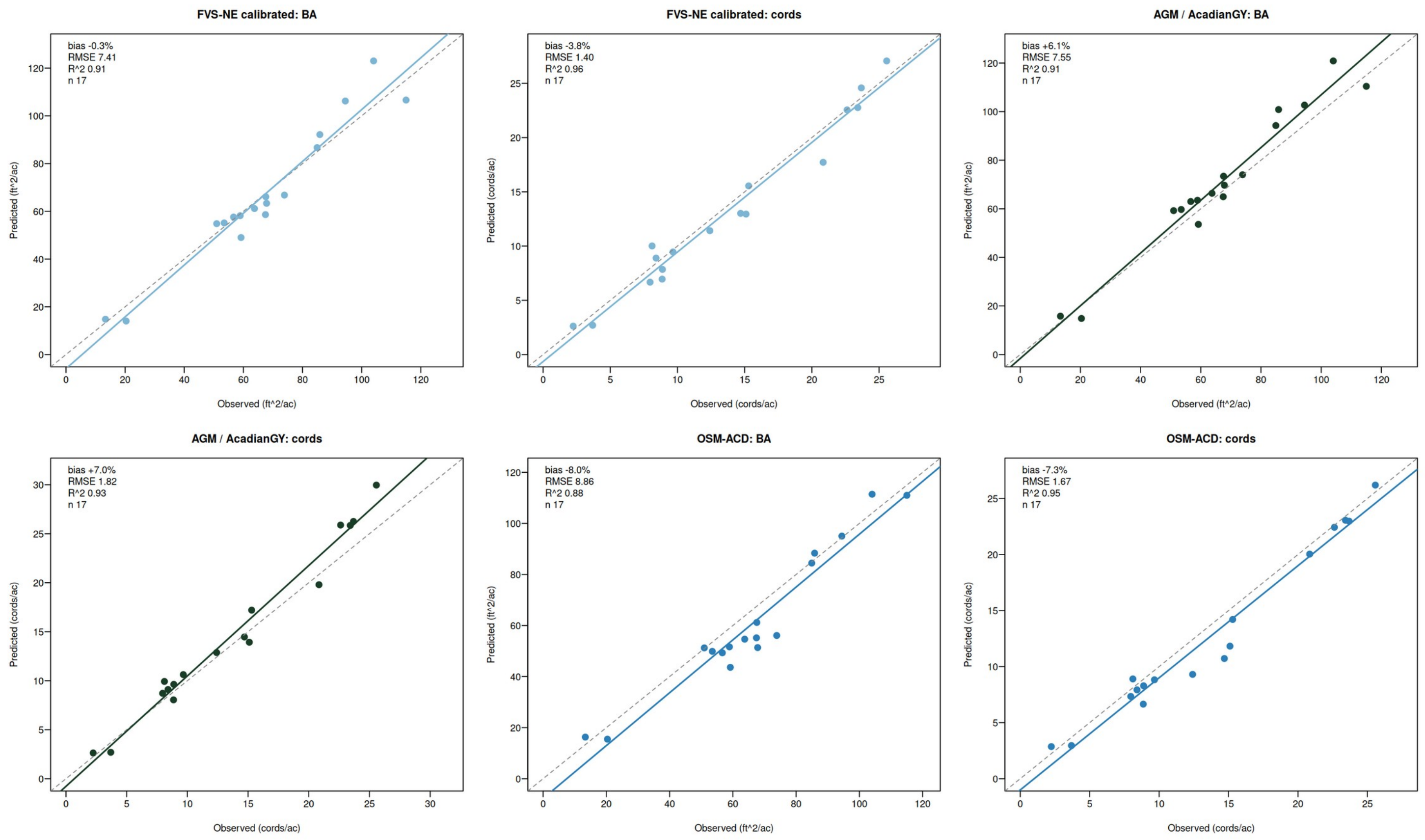
Cedar / A+B is uniquely covered by CFI but NOT by AcadianGY byStrata — the two anchors are complementary

Source: SILC_CFI_FIADB_Database_v16.xlsx, FIADB-structured. AcadianGY MORTCAL Cardinal job 11105908.

v16 expansion: 9× larger empirical anchor across 154 SILC plots

Three models bracket the truth on CFI

FVS-NE calibrated lands closest, AcadianGY over projects, OSM-ACD under projects



AcadianGY 12.3.9 (in-source mortality + ingrowth fix); OSM-ACD = Open Stand Model Acadian v2.26.1, Cardinal SLURM 10988107 + 10990880

FVS-NE calibrated and AcadianGY near zero, OSM is the under bracket

Multi model CFI accuracy at a glance

Routine growth subset, n=17. Best of each metric in green, worst in red

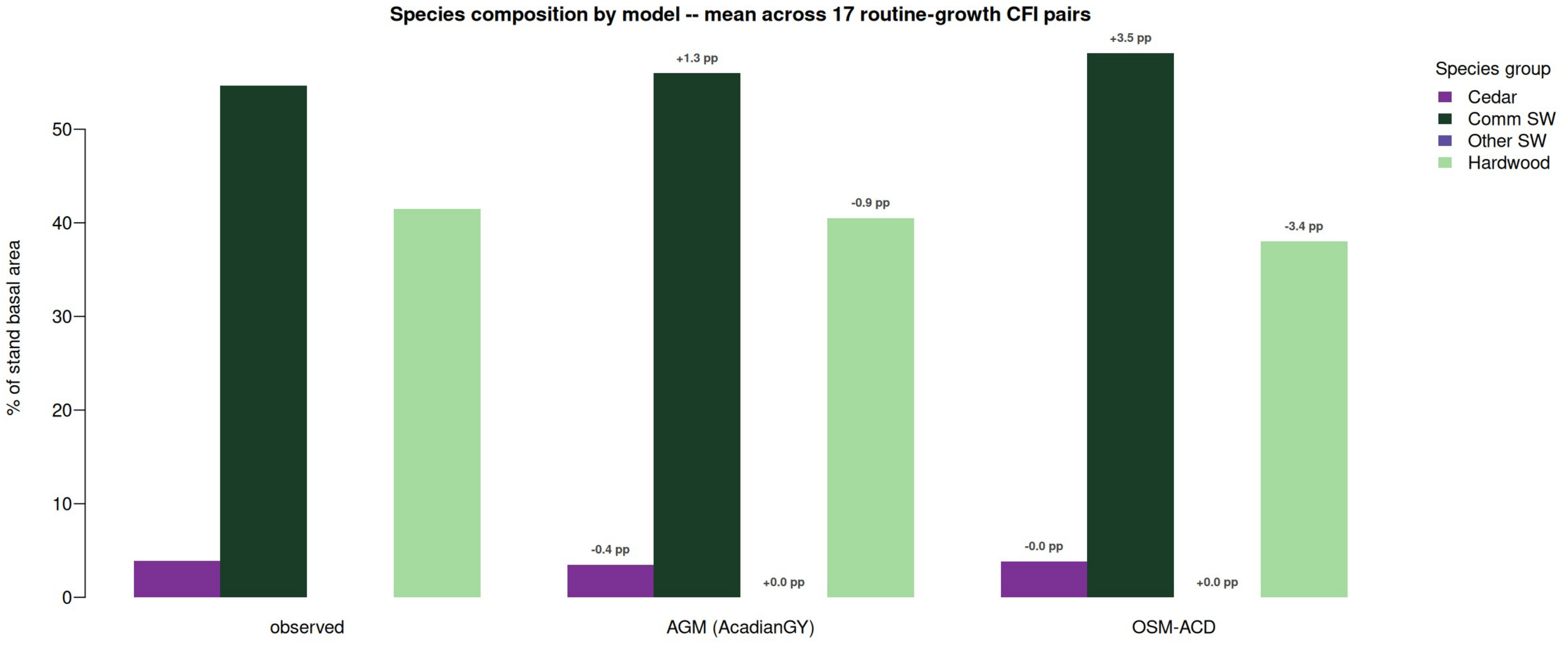
Predictor	BA	Cords	BdFt (Intl ¼)	SDI	Curtis RD
zero growth	-1.8%	-6.0%	-15.9%	+4.8%	+5.2%
FIA prior PAI	+3.9%	+1.4%	-8.9%	n/a	n/a
AcadianGY 12.3.9	+6.1%	+7.0%	**+8.4%**	**-0.3%**	**-0.6%**
FVS-NE default	+4.6%	-1.3%	-11.9%	-1.0%	-1.1%
FVS-NE calibrated	**-0.3%**	-3.8%	-13.6%	-5.9%	-6.1%
FVS-ACD (default = cal *)	+5.3%	**-2.5%**	-11.5%	-0.5%	-0.7%
OSM-ACD	-8.0%	-7.3%	-6.6%	+2.0%	+6.0%

** FVS-ACD calibration tweaks BAMAX stocking caps which do not bind below 150 ft^2/ac BA; default and calibrated produce identical projections on these CFI plots. n=17 routine-growth subset*

FVS-NE calibrated wins BA, FVS-ACD wins cords, AcadianGY wins BdFt SDI RD

Species composition matched by AcadianGY and OSM-ACD

AcadianGY within 1.3 pp on Comm SW share, OSM-ACD within 3 pp on all four groups



Mean of routine-growth subset n=17. Bias annotations are predicted minus observed in percentage points. FVS per tree species output deferred (TREELIST scheduling)

AcadianGY and OSM-ACD both track the observed Comm SW + Hardwood mix

What this means for SILC operational use

Benchmark SILC AGM on the same CFI and byStrata data — the missing model

AcadianGY is the CRSF R model in this deck. SILC's operational AGM has NOT been run on the SILC CFI pairs or on the same 11 byStrata stands. The clearest single next step is for SILC to share AGM trajectories on these stands so the four model picture becomes a five model picture with the operational anchor included.

AcadianGY with MORTCAL is the closest CRSF analog for an operational anchor

Cuts 10 yr empirical BA bias from +18.5% to +7.8%; year 100 BA settles at 98 to 108 ft²/ac, Cords 35 to 39 cords/ac. Sawlog BdFt R^2 = 0.84 with single digit bias. Until SILC AGM is benchmarked, use AcadianGY MORTCAL as the operational floor.

FVS-NE calibrated brackets the upper bound on year 100 and is best on short horizon

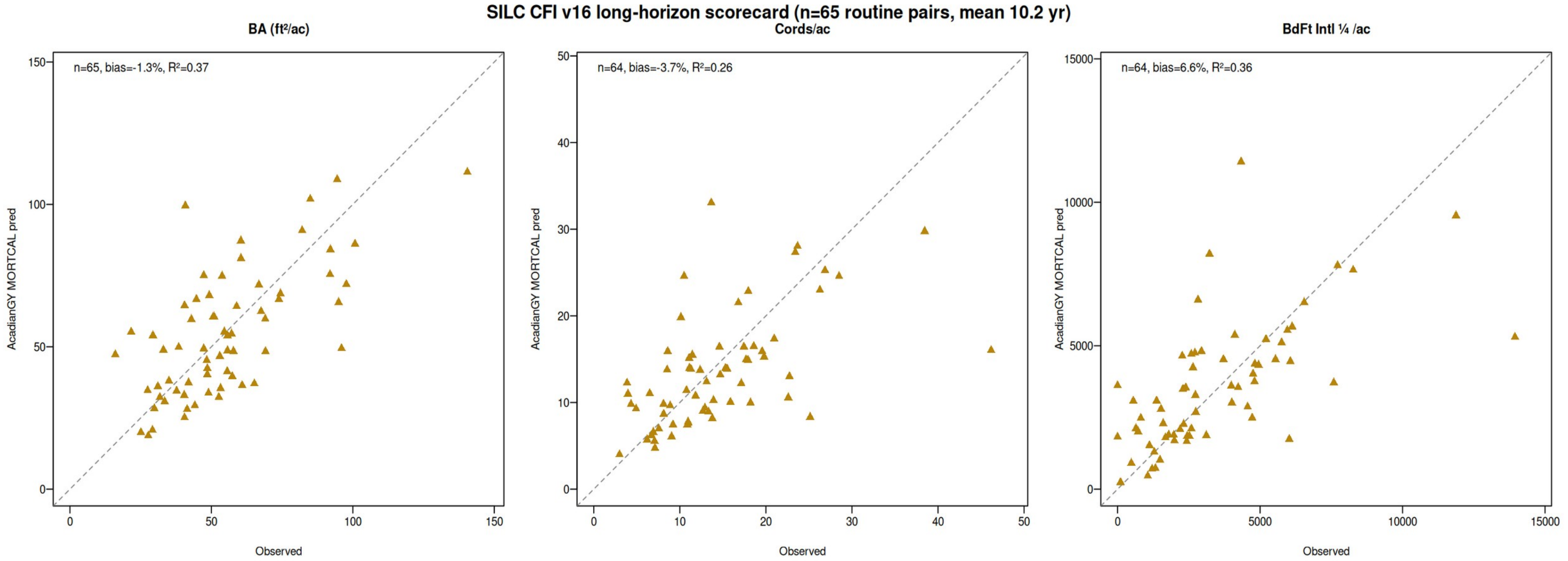
Year 100 BA 125 to 225 ft²/ac, Cords 44 to 70. Sits between AcadianGY default and MORTCAL on every cell. Lowest 5 yr CFI bias (-0.3% BA, -3.8% Cords). Use as the FVS variant cross check.

OSM ACD as the additional bracket

SILC AGM benchmark is the missing piece; AcadianGY MORTCAL is the closest CRSF analog

Long horizon scorecard expanded to n=65 plots (v16 CFI)

SILC shared CFI v16 with 154 plots (was 10 in v3). The empirical anchor is now ~9× larger. AcadianGY MORTCAL on 65 routine pairs.



Model	n	BA bias	BA RMSE	BA R^2	Cords bias	Cords R^2
AcadianGY MORTCAL	65	−1.3%	18.1	+0.37	−4.7%	+0.27
OSM-ACD	63	−2.6%	18.9	+0.33	n/a	n/a
FVS-NE default	65	+3.1%	20.0	+0.24	−4.4%	+0.20
FVS-ACD default	65	+3.2%	20.0	+0.24	−5.3%	+0.20
FVS-NE calibrated	65	−7.1%	18.1	+0.37	−12.1%	+0.23

n=65 routine pairs (8 establishment events excluded). Cardinal jobs 11105908 (AcadianGY), 11106519 (FVS), 11106515 (OSM).

AcadianGY MORTCAL lowest BA bias (−1.3%); OSM-ACD second; all 4 models within ±7% on n=65

CFI scorecard fits a larger validation picture

n=17 SILC CFI pairs anchored against 12029 plot Maine FIA validation

17	12,029	-0.06%
SILC CFI routine remeasurement pairs	**Maine + NH + VT FIA paired plots**	**FVS-ACD calibrated BA bias on FIA**
Davistown, ME 1981 to 2000	*Apples-to-apples overstory recompute, DIA >= 5 in*	*Independent of CFI scorecard, n 700x larger*

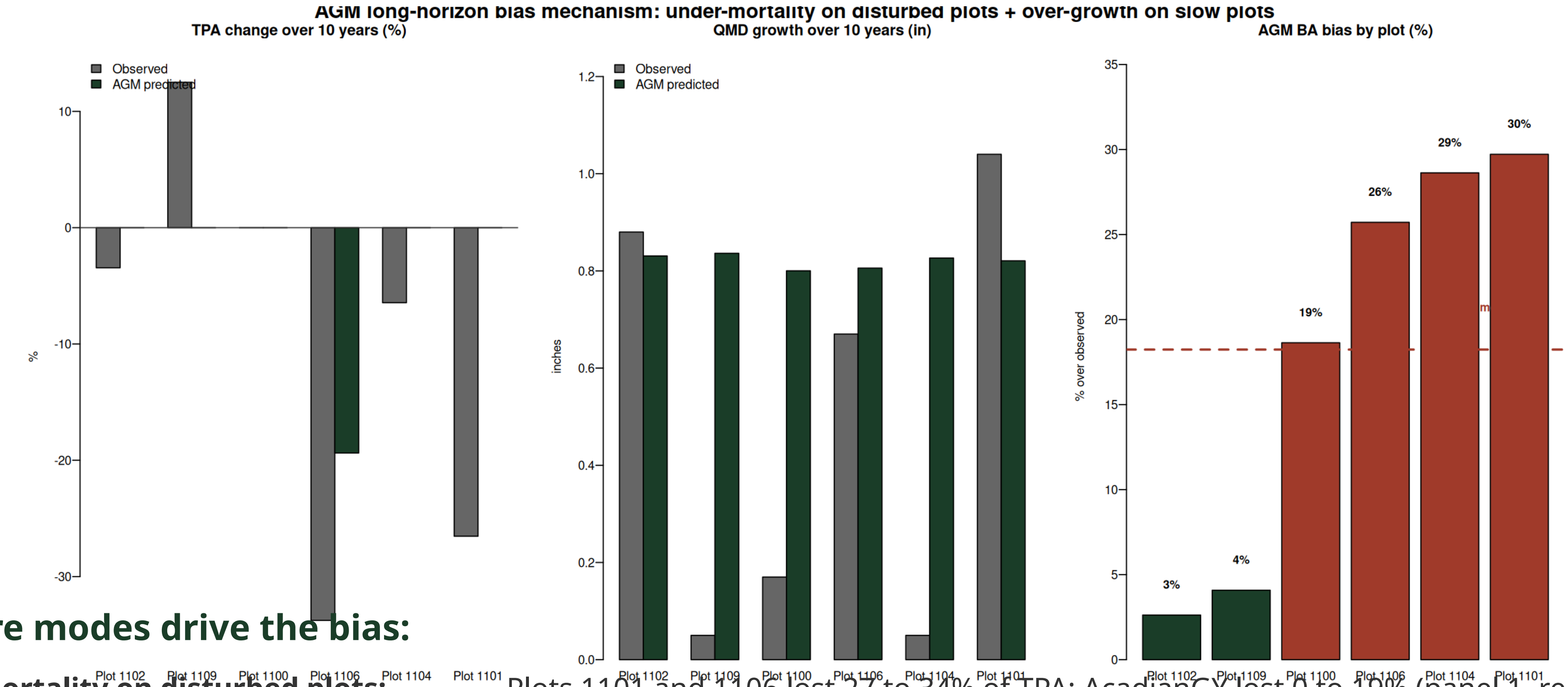
The +5% to +6% AcadianGY and FVS-ACD CFI biases are consistent with the +0% bias the same calibration achieves on 12,029 FIA plots when stand structure differences are accounted for. The CFI biases reflect SILC's Acadian Matrix configuration, not a model failure.

FIA recompute: SLURM 10591333; calibration/output/comparisons_overstory/FINDINGS.md

Small n CFI scorecard is consistent with the large n FIA validation

Where the +18.5% AcadianGY bias actually lives

Two mechanisms drive the long-horizon over-projection: under-mortality + uniform DBH growth.



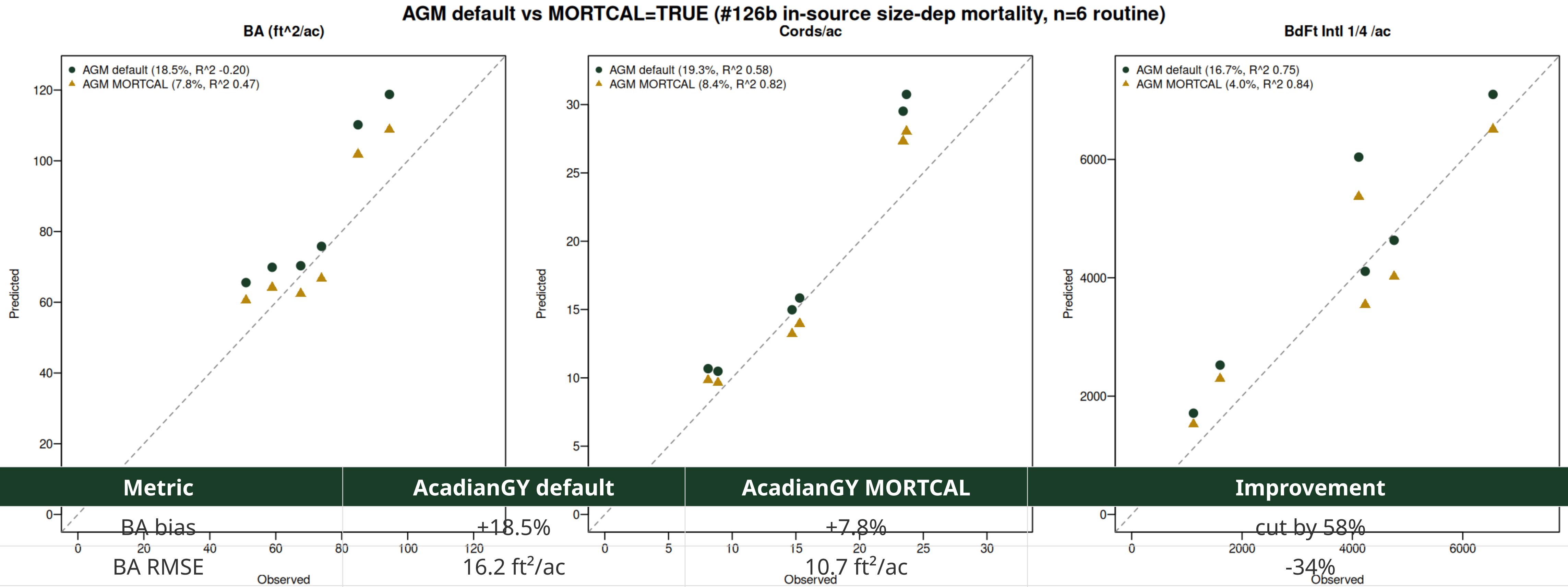
Two failure modes drive the bias:

- **Under mortality on disturbed plots:** Plots 1101 and 1106 lost 27 to 34% of TPA; AcadianGY lost 0 to 19% (panel 1, red bars in panel 3).
- **Over growth on slow plots:** AcadianGY grew QMD a uniform +0.8 in regardless of site; plots 1100 and 1104 grew only 0.05 to 0.17 in observed (panel 2).
- **Verified, not a calculation issue:** BA / Cords / BdFt pipeline reproduces from the AcadianGY treelist to 1e-13 ft^2/ac. The bias is

Bias is mechanism, not bug: under mortality + over growth on edge cases

Turning on AcadianGY MORTCAL cuts long horizon bias by more than half

#126b size-dependent mortality patch (Maine FIA calibration). Default OFF in 12.3.9; enable via ops$MORTCAL=TRUE.



Metric	AcadianGY default	AcadianGY MORTCAL	Improvement
BA bias	+18.5%	+7.8%	cut by 58%
BA RMSE	16.2 ft²/ac	10.7 ft²/ac	-34%
BA R^2	-0.20	+0.47	flips negative to positive
Cords R^2	0.58	0.82	+0.24
BdFt bias	+16.7%	+4.0%	down to single digit

AcadianGY with MORTCAL is the new long horizon champion across BA, cords, sawlog BdFt

Next steps

1. **Confirm the 5×2 stratification with SILC**
   Sign off that the species composition rules and density threshold match your operational categories

2. **Backfill the four empty cells with plot level AcadianGY MORTCAL**
   Cedar/A+B, Mixedwood/A+B, both Other Softwood; pull from individual SILC CFI plots assigned to those cells, run AcadianGY MORTCAL 100 yr

3. **Validate year 100 BA against SILC operational expectations**
   MORTCAL 100 yr BA settles at 80 to 110 ft^2/ac. Confirm this matches SILC's expected carrying capacity for managed Acadian stands

4. **Set the uncertainty bands and ship the operational handoff**
   Use AcadianGY MORTCAL RMSE 10.7 ft^2/ac BA at 10 yr as the per plot uncertainty. Year 100 cords/ac estimates: 36 to 39 across strata

Four model scaffold is live, awaiting SILC sign off

Questions and discussion

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